Can Machine Learning Variable Selection and Individual Commodity Fundamentals Improve Futures Return Forecasting?

Leo Specks
Supervisor: Prof. Massimo Guidolin
MSc Finance, Università Bocconi

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Figure 1
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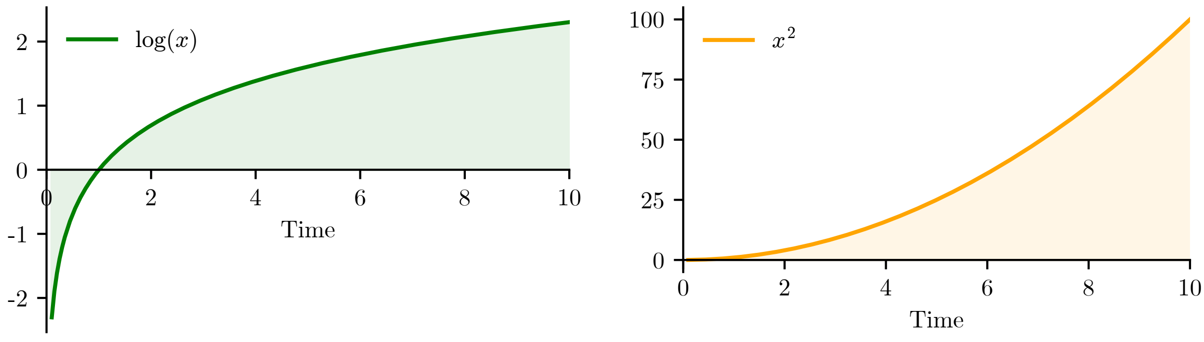

(a) Logarithm Function

(b) Quadratic Function

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Figure 2

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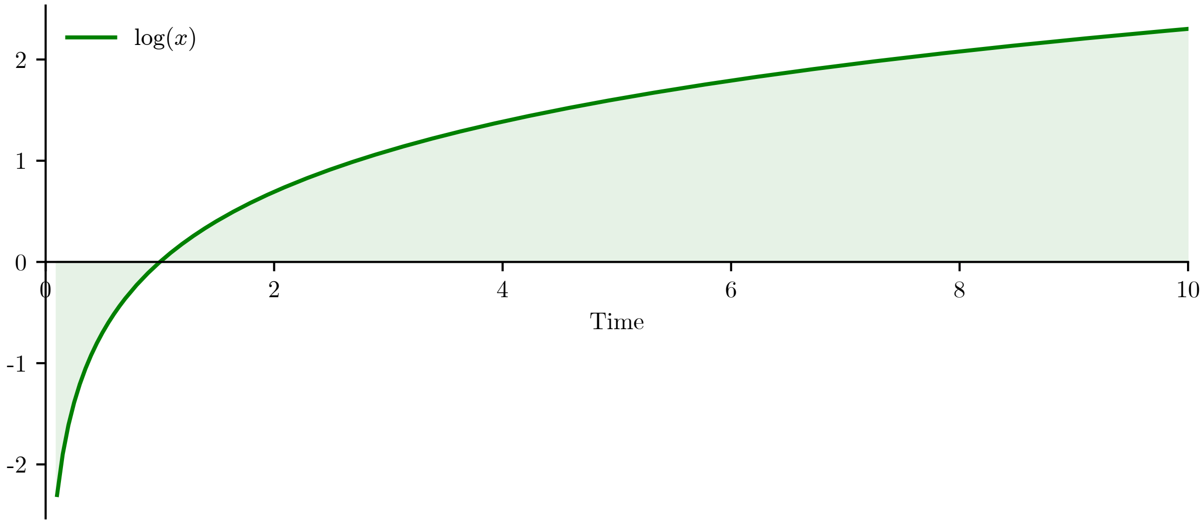

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Table 1
UMD Factor: Summary Statistics (Jan 1927–Feb 2026)

Statistic	Monthly	Annualized
Mean Return (%)	0.612	7.34
Standard Deviation (%)	4.68	16.21
Sharpe Ratio	-	0.453
t-statistic	4.51	-
p-value (two-tailed)	7.11×10^{-6}	-
Number of Months	1,190	-

tincidunt eget. Praesent dictum sed eros in pharetra. Vivamus nec dictum risus. Integer ac massa at magna rutrum pulvinar eget nec nunc. Curabitur tempor at lorem vitae venenatis. Suspendisse auctor posuere iaculis. Suspendisse egestas lobortis diam. Integer mattis. The approach by Guidolin and Pedio (2021) has shown that this is realistic[1].

10.2 Citations

This is a cite: Guidolin and Pedio, 2021, this a textcite: Guidolin and Pedio (2021), this a parencite: (Guidolin & Pedio, 2021), this a citeauthor: Guidolin and Pedio, this a citeyear: 2021, and this a parencite with two references: (Giampietro et al., 2018; Guidolin & Pedio, 2021).

Table 2
Citation Styles with BibLaTeX

Command	authoryear	numeric
`\cite{}`	Author, Year	[number]
`\textcite{}`	Author (Year)	Author [number]
`\parencite{}`	(Author, Year)	[number]
`\citeauthor{}`	Author	Author
`\citeyear{}`	Year	Year (not linked)

Use `\parencite{}` at the end of a statement and `\textcite{}` when referencing in the middle of a sentence.

10.3 Codeblock

The algorithm can be viewed in Codeblock 1

[1] This would be the first footnote.

Code 1
Example Codeblock

```
# KPI OVERVIEW
kpis = pd.DataFrame({
    "Ann. Return (approx)": ann_return_simple,
    "Ann. Return (compounding)": ann_return,
    "Ann. Volatility": ann_vol,
    "Sharpe": sharpe,
    "CAGR": cagr,
    "Max Drawdown": max_dd
})

kpis.T
```

References

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Appendix